Assignment 5

Group 7: Fina, Jenrik, David, Jesper

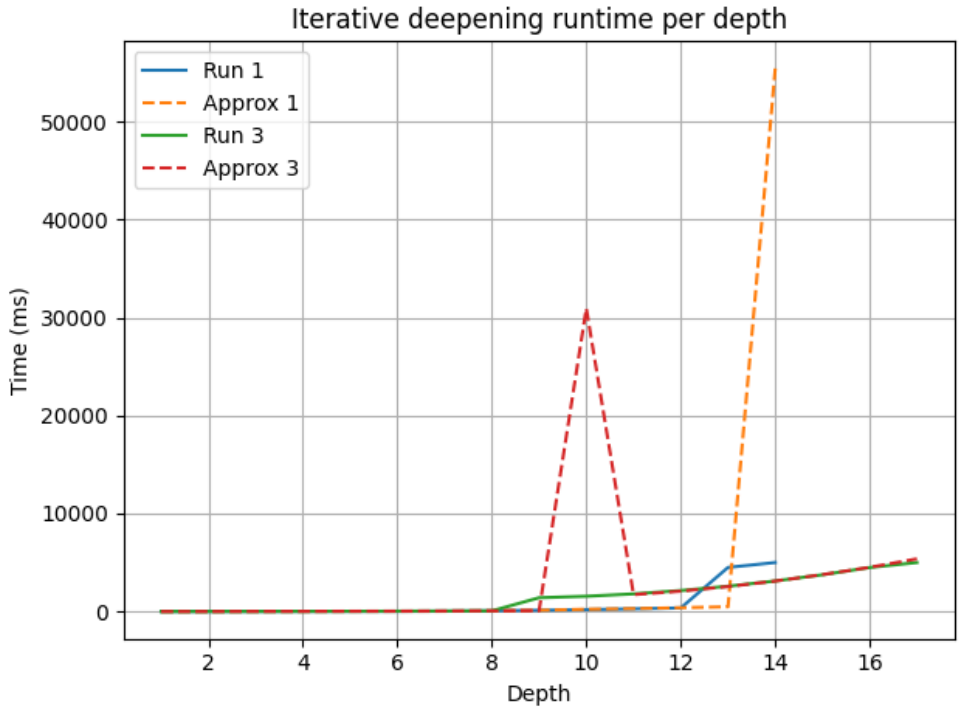
Task 1: Estimating the required time

Use measurements of previous depths

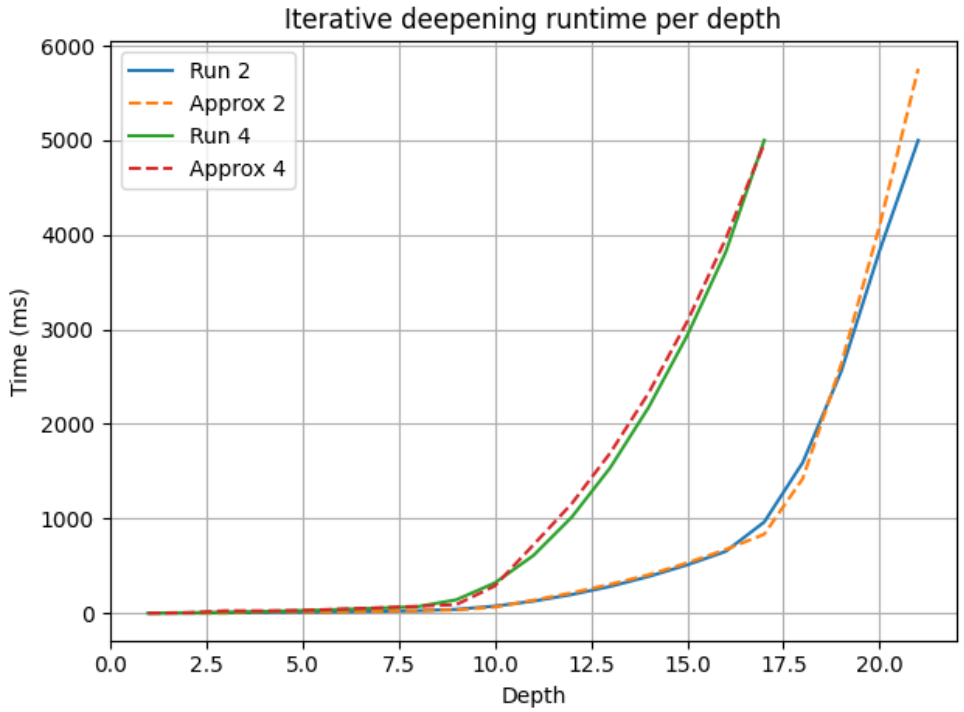
For each depth n d_n:

1. Calculate ratio $r = \frac{d_{n-1}^2}{d_{n-2}}$
2. Take maximum of max_ratio and current ratio
3. Use estimate to decide whether further iteration

Task 1: Estimating the required time



Estimate without max_ratio

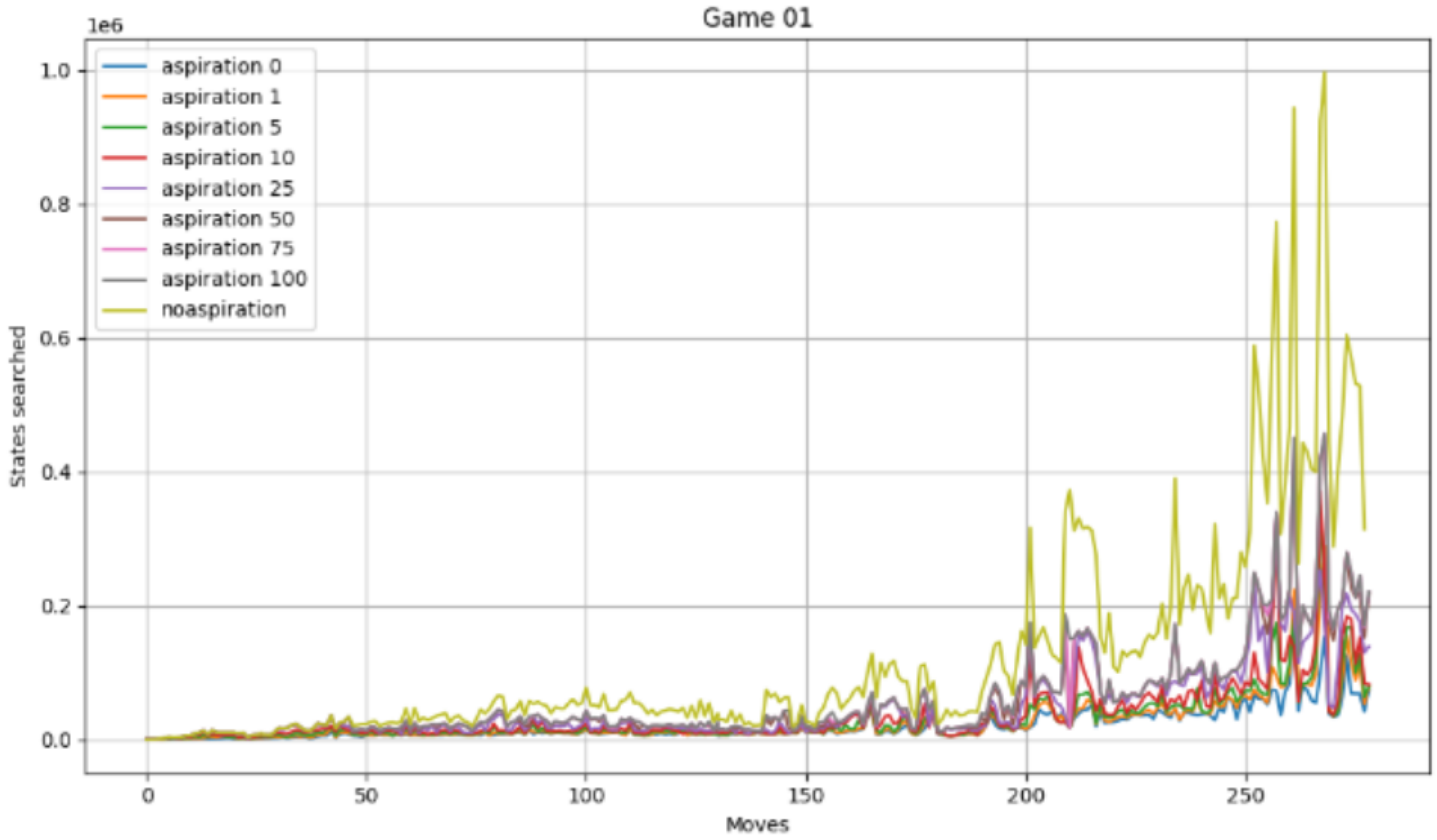


Estimate with max_ratio

Task 2: Implementing Aspiration Windows

- aspiration_window_ (win) guess for the middle point of the window, set to previous top_score
- aspiration_delta_ (Δ) constant delta for the window (distance from middle point to Edge)
- Run Alpha-beta prunning, with α= win - Δ and β = win + Δ
- If the search fails, rerun with one-sided full-width window

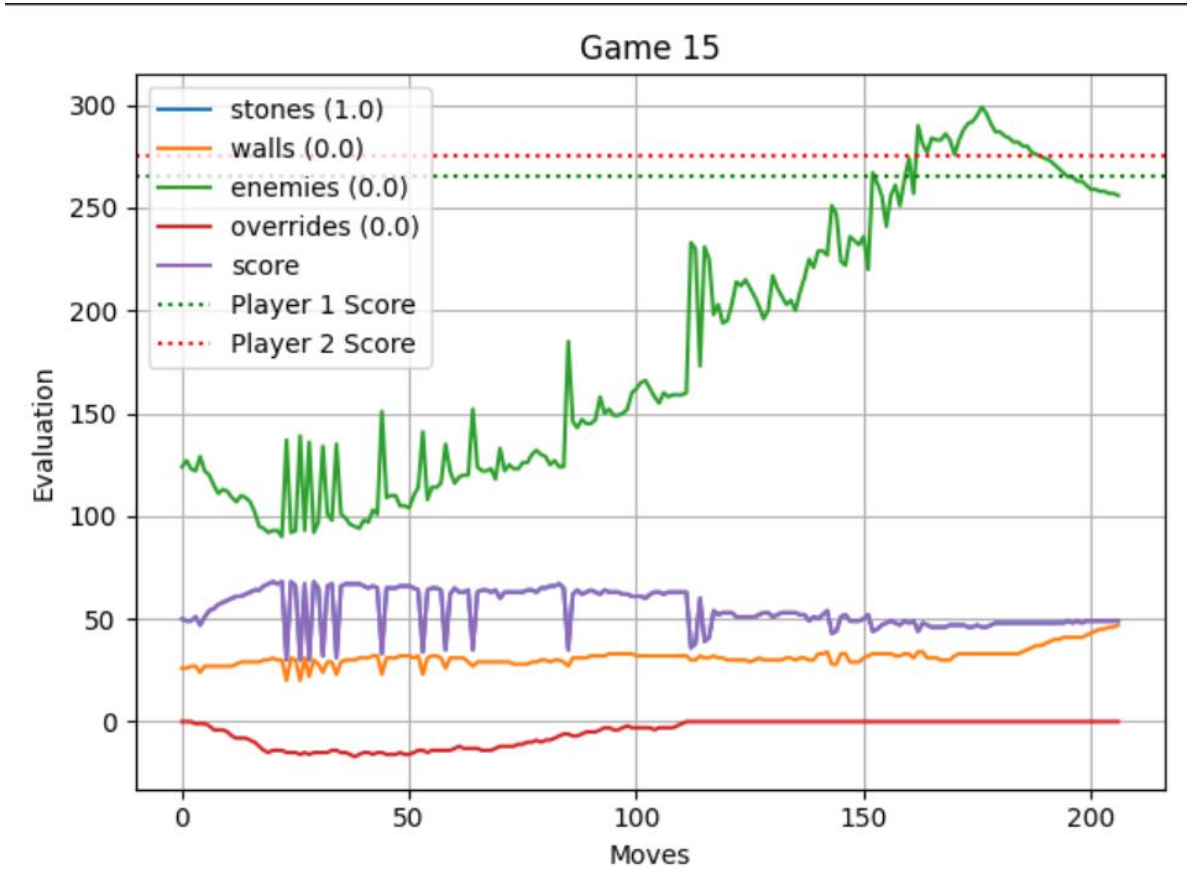
Task 3: Aspiration Window Analyses



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- Size 0 saves avg. 92.57%
- Size 5 saves avg. 87.86%
- Size 50 saves avg. 65.21%
- Size 100 saves avg. 52.98%

Task 4: Heuristic Analyses recap



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- Override stones : 5.009541
- Stone count : 4.314854
- Adjacent opponent : 2.49639
- Adjacent borders : 4.770165
- Mobility : 1.900653
- Last move : 1.15138